Harnessing Bayesian-Filter-Derived SFR Method Considering Uncertain-variability Features in Refined FCUC

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***Abstract*—Frequency dynamic fills with uncertain-variability features due to the increasing penetration of converter-interfaced generators, where this feature trends to be bottlenecks in selecting scheduled commitments solved by Frequency-Constrained Unit Commitment(FCUC). To address this, a Bayesian-filter-derived System-Frequency-Response(SFR) model is proposed and a refined FCUC(r-FCUC) is extended. Significantly, the uncertain-variability feature due to the intermittent output of converter-interfaced generators, PMU measured error, etc., is treated as a chronological stochastic process. As such, the frequency dynamic can be extracted as a sequential deterministic process within a Bayesian-filter framework, on this basis, an improved Bayesian-Filter-derived SFR model is established. In contrast to existing methodologies such as analytical SFR or time-discrete SFR methods, the proposed method can capture the trajectory of frequency dynamics in the uncertain-variability environment and estimate its posterior distribution. Furthermore, the frequency dynamic via the proposed SFR method is reshaped as a set of chance-constrained frequency dynamic constraints and subsequently integrated into r-FCUC. Since reformulated frequency dynamic constraints in r-FCUC consider uncertain-variability features, the calculated scheduled commitment through r-FCUC can well require actual frequency requirements. Several results on a modified IEEE 6 bus test system and IEEE 118 bus test system are implemented to check the validity of the proposed methods.**

***Index Terms*—Uncertain-variability, Bayesian filter, system frequency response, converter-interfaced generators, frequency-constrained unit commitment.**

I. INTRODUCTION

INCREASING penetration of renewable energy challenges the security of power systems under complex disturbance cases. In such situations, independent system operators (ISO) must set aside sufficient resources to guarantee the security of scheduled commitments. Nevertheless, since SFR has significant uncertain-variability features along with non-fluctuating controllable generators replaced by converter-interfaced ones, it is hard to maintain the security of scheduled commitments without considering uncertain-variability features. Therefore, developing a reliable SFR-constructed method and associated FCUC model is essential to capture the uncertain-variability features of frequency dynamic.

Existing literature has reported a amount number of analytical models and time-discretized models to analysis frequency dynamics. The analytical models reflect the average system frequency changing among all synchronize generators. It posits that those generators with similar control characteristics will maintain standard frequency dynamics. Consequently, the generators participating into SFR can be aggregated into a single unit or a limited number of units, which further simplifies the swing function and prevents non-analytical solutions. Ref 1 proposed a second-order swing function to represent the frequency dynamics of systems dominated by conventional generators. Ref 2 introduced a refined method for identifying frequency parameters in low-inertia grids and extended the analytical SFR model. Ref 3 integrated the linearized control schemes of converter-interfaced generators into frequency dynamics and then expanded a two-machine SFR method. These analytical models yield similar results, with the impact of heterogeneous generators on SFR primarily dependent on the selection of local frequency regulation parameters. Another analytical model, known as average frequency response, assumes that different generators have specific control characteristics while maintaining consistent frequency dynamics, as discussed in Ref. 4. Although this ASF modeling method may be more accurate, it often requires an analytical solution.

Conversely, the discretized model offers greater flexibility and effectiveness regarding multi-machine frequency dynamic modelling and various control characteristics analysis. These models utilize interpolation methods, such as the Euler method and the Runge-Kutta method, to solve the differential equations of frequency dynamics. Ref 4 assumed that reheated generators share similar time-delay characteristics and derived a discretized solution for frequency dynamics. Ref 5 considered different time-relay characteristics of converters and developed a corrected discretized method. Ref 6 incorporated the droop control characteristics of flywheel storage into frequency dynamics and constructed a discretized model. Ref 7 comprehensively accounted for the effects of converter-interfaced generators alongside conventional generators and rebuilt a discretized model. These discretized model highlight the evolution of frequency dynamics while considering various control schemes. Nevertheless, they requires more computing resource in comparison with the analytical models. In terms of the aforementioned SFR models, the analytical SFR models are more suitable for the analysis of frequency changing in the dynamic term, while the discretized SFR ones are broadly used to identify the impact of various control schemes on frequency dynamics. In addition, the uncertain-variability feature of converter-interfaced generators participating in frequency support has not yet been thoroughly examined

On the overall SFR horizon, the uncertain-variability feature caused by converter-interfaced generators and measured errors is becoming a significant factor that affects frequency dynamics, and this issue is pronounced in the low-inertia grid. The reason for this issue is that converter-interfaced generators become increasingly indispensable, as conventional generators cannot fully provide adequate frequency reserves. As a consequence, the uncertain-variability feature are bound to have an impact on the frequency dynamics itself. Ref 8 analyzed frequency dynamics by considering uncertain-variability features as a $It\hat{o}$ processes, where uncertain-variability features are retreated to be a prior sequential random sequence following

Gaussian distribution. Due to excellent conjugate properties of Gaussian distribution families, the SFR derived from the $It\hat{o}$ process result in an analytical solution. Nevertheless, the factors contributing to uncertain-variability features typically depend on frequency variations in the actual engineering process. Thereby, the distribution of uncertain-variability features is conditional and unknown. From this standpoint, existing models in [ ] addressing uncertain-variability factors cannot accurately reflect the reliable uncertain-variability factors related to frequency dynamics.

Furthermore, the scheduling plan solved by FCUC should be checked to ensure that the system have capabilities to satisfy frequency security requirements. Generally, frequency security requirements consist of rate-of-frequency-of-change (RoCoF), frequency nadir, and quasi-steady-state frequency derivation (Qss); they are usually reshaped as a group of linear or nonlinear constraints. Ref.7 constructed a set of nonlinear RoCoF, nadir, and Qss constraints derived from analytical SFR model and developed a FCUC. Ref.8 constructed a max-affine-derived linear frequency nadir constraint and constructed a refined FCUC. Ref.9 applied an active linearized sparse neural network to learn mapping relationships between frequency nadir and frequency parameters. Ref.10 designed a most-miniature-square method to estimate the frequency parameters of frequency nadir constraints. Ref. 10 developed a discretized frequency derivation constraints. while reformulated frequency constraints in FCUC models are deterministic, the uncertain-variability feature throughout the frequency dynamic processes is implicitly ignored.

Motivated by such concerns, this study constructs a new type of SFR model considering uncertain-variability features. Note that uncertain-variability features can be forecasted as stochastic waveform signals using PMU, so the posterior distributions of uncertain-variability features can be directly calculated. Based on this, both trajectory and distribution of frequency dynamic are represented as a sequential stochastic process, soluted by the Bayesian filter. Subsequently, the probabilistic characteristics of frequency dynamic under uncertain-variability features are captured and then reformulated as chance-constrained frequency constraints into FCUC. Through this work, the probabilistic characteristics of frequency dynamics caused by uncertain variability features can be efficiently examined through the modeling of SFR and FCUC.

To summarize, two significant contributions in this work are described below, 1) A new SFR model considering uncertain-variability features is constructed, and a associated solution method within Bayesian filter theorem is carried out. Using this model, both changing and probabilistic distribution of frequency dynamics can be estimated. 2) In connection with the proposed SFR model, a new chance-constrained formulation related to frequency dynamic constraints is introduced, along with the development of a corresponding FCUC.

The remainder of this paper is organized: Section 2 introduces a Bayesian-filter-derived SFR model. Section 3 constructs chance-constraint frequency dynamic constraints. 4 extends the formulation of chance-constraint FCUC. Section 5 verifies the effectiveness of the proposed model in a modified IEEE 14 bus test system and IEEE 118 bus test system, and Section 6 summarizes the primary findings and conclusions presented in this research.

II. FREQUENCY DYNAMIC MODEL CONSIDERING UNCERTAIN-VARIABILITY FEATURES

This section examines the impact of uncertain-variability features on frequency dynamics and then develops a Bayesian filter-based SFR model. Specifically, it derives the posterior distribution of frequency dynamics using a Bayesian filter and outlines the calculation process using a sequential importance sampling (SIS) algorithm.

Mathematically, the transfer function of SFR concerning a occurred step-type power disturbance $\Delta p_{\text{step}}(s) = \Delta p_{step}/s$ can be rewritten as,

$$\begin{aligned}(M_G s + D_G)\,\Delta f(s) = (1/s\Delta p_{\text{step}}) - \\ \left(K_G R_G^{-1}\right)\left[(1+F_G T_G s)/(1+T_G s)\right]\Delta f(s) - \\ \left(K_W R_W^{-1}\right)\left[(H_W + D_W)/(1+T_W s)\right]\Delta f(s) - \\ \left(R_W^{-1}\right)\left[1/(1+T_W s)\right]\Delta f(s) - \left(R_E^{-1}\right)\left[1/(1+T_E s)\right]\Delta f(s)\end{aligned} \tag{1}$$

The time domain expression of (1) can be transferred by utilizing inverse-Laplace-transformation (ILT),

$$2H_\Sigma \Delta f'(t) + D_\Sigma \Delta f(t) = \Delta p(t) \tag{2a}$$

$$\Delta p(t) = \Delta p_{step} - (\Delta p_G(t) + \Delta p_{W,1}(t) + \Delta p_{W,2}(t) + \Delta p_E(t)) \tag{2b}$$

$$H_\Sigma = H_G + H_W,\, D_\Sigma = D_G + D_W + R_W^{-1} + R_E^{-1} \tag{2c}$$

$$\begin{aligned}\Delta p_G(t) = K_G F_G R_G^{-1}\Delta f(t) - \\ K_G(1-F_G)R_G^{-1}e^{-\frac{t}{T_G}}\int_0^t e^{\frac{\kappa}{T_G}}\Delta f'(\kappa)d\kappa\end{aligned} \tag{2d}$$

$$\Delta p_{W,1}(t) = \left(K_W R_W^{-1}\right)\Delta f(t) + \left(2H_W \Delta f'(t) + D_W \Delta f(t)\right) \tag{2e}$$

$$\Delta p_{W,2}(t) = \left(R_W^{-1}\right)\Delta f(t),\, \Delta p_E(t) = \left(R_E^{-1}\right)\Delta f(t) \tag{2f}$$

where (2a) and (2b) describes the swing function; (2c) denotes aggregated frequency parameters; (2d)-(2c) define additional rescheduled output of different generators. By substituting $\Delta f'(t) = {}^{[\Delta f(t) - \Delta f(t-\delta t)]}/{}_{\delta t}$ into (2a) and (2d), the relations among $\Delta f(t), \Delta p_G(t)$ and $\Delta p_{W,1}(t)$ between adjacent periods can be estimated,

$$\Delta f(t+\delta t) = \left[0.5H_\Sigma^{-1}\Delta p(t) - \alpha_1 \Delta f(t)\right]\delta t \tag{3a}$$

$$\Delta p_G(t) = K_G R_G^{-1}\Delta f(t) - \alpha_2 \int_0^t e^{\frac{t}{T_G}}\left[\Delta f(t+\delta t) - \Delta f(t)\right]dt \tag{3b}$$

$$\Delta p_{W,1}(t) = \alpha_3 \Delta f(t+\delta t) - 2H_W \delta t \Delta f(t) \tag{3c}$$

$$\alpha_1 = 0.5 D_\Sigma H_\Sigma^{-1} - 1 \tag{3d}$$

$$\alpha_2 = K_G(1-F_G)R_G^{-1}e^{-\frac{1}{T_G}t} \tag{3e}$$

$$\alpha_3 = K_W R_W^{-1} + D_W + 2H_W \delta t \tag{3f}$$

It can be observed that $\Delta f(t+\delta t)$ is close to $\Delta p(t)$ while all generators are automatically adjusting their additional outputs so as to eliminate the imbalanced power disturbance. Despite that $\Delta p_G(t), \Delta p_{W,1}(t)$ in (3b) and (3b) can be recursively solved after observing $\Delta f(t+\delta t)$ and $\Delta f(t)$, the uncertain variability feature has not yet introduced. To address this, a corrected solution regarding $\Delta f(t+\delta t)$ that considers uncertain variability features is categorized. The uncertain-variability feature contains two parts; the first part is caused by unlimited restriction of additional outputs $\langle \Delta p_G(t), \Delta p_{W,1}, \Delta p_{W,2}\rangle$, it means that the additional output of generators who provide frequency support under SFR cannot be reached due to their maximum outputs restriction. Unforeseeable uncertain output characteristics of converter-interfaced generators, as well as possible measured errors cause the second part. Let $\Delta\omega_1(t) = -(\lceil \Delta p_G(t), \overline{p}_G \rceil + \lceil \Delta p_{W,1}(t), p_{W,1}(t) \rceil + \lceil \Delta p_{W,1}(t), p_{W,2}(t) \rceil)$ enforces the corrected term for preventing generators from exceeding their maximize outputs, $\Delta\omega_2(t) = \mathrm{Normal}(t;\mu,\sigma)$ describes uncertain fluctuations of converter-interfaced generators, thus, uncertain-variability features can be estimated by adding $\Delta\omega_1$ and $\Delta\omega_2$ together,

$$\Delta f(t+\delta t) = \left\{0.5H_\Sigma^{-1}[\Delta p(t)+\Delta\omega_1+\Delta\omega_2]-\alpha_1\Delta f(t)\right\}\delta t \quad (4a)$$

In (4), the above equation does not keep a closed analytical solution because $\Delta\omega = \Delta\omega_1+\Delta\omega_2$ doesn't belong to any prior distribution, therefore, the Bayesian filter is applied to address it. Let $\mathcal{P}(\Delta f(t))$,$\mathcal{P}(\Delta p(t))$ enforces the posterior of $\Delta f(t)$ and $\Delta p(t)$, $\Delta f(t) \Rightarrow \Delta f(t+\delta t)$ can be retreated as a hidden decision process, while $\Delta p(t) \Rightarrow \Delta p(t+\delta t)$ be an observable process. Consequently, $\Delta f(t+\delta t)$ can be solved alternately, as illustrated in figure.(2).

$$\mathcal{P}(\Delta f(t+\delta t)|\Delta p(t_0:t+\delta t);\Delta\omega) = \overbrace{\mathcal{P}(\Delta p(t+\delta t)|\Delta f(t+\delta t);\Delta\omega)}^{\text{likelihood function}} \underbrace{\mathcal{P}(\Delta f(t+\delta t)|\Delta p(t_0:t);\Delta\omega)}_{\text{prior function}} \mathcal{P}^{-1}(\Delta p(t+\delta t)|\Delta p(t_0:t);\Delta\omega) \quad (5a)$$

where the likelihood in (5a) describes relation between $\Delta p(t+\delta t)$ and $\Delta f(t+\delta t)$; prior function describes sequential recursive processes concerning frequency dynamic from $\Delta f(t_0:t)$ to $\Delta f(t+\delta t)$. Using the prior and predicated processes within the Bayesian inference framework, the impact of uncertain-variability features, denoted by $\Delta\omega$, can be well incorporated. The expression of the prior function in (5a) can be recursively estimated as follows,

$$\mathcal{P}\left(\Delta f(t)\,|\,\Delta p\left(t_0:t+\delta t\right),\Delta\omega\right) = \int [\mathcal{P}(\Delta f(t+\delta t)\,|\,\Delta f(t_0:t),\Delta\omega)\mathcal{P}(\Delta f(t)\,|\,\Delta p(t_0:t)\,,\Delta\omega)]d\Delta f(t) \quad (6a)$$

Both probabilistic functions in (5a) and (6a) can not be directly solved because their distributions are unknown, hence, the sequential important sampling algorithm (SIS) is applied to address this issue. Let $\mathcal{Q}(\Delta f(t+\delta t)|\Delta p(t_0:t+\delta t);\Delta\omega)$ denote the importance distribution, which can be treated as a replaceable proposal distribution for generating samples (filters); $\lambda(\Delta f(t_0:t+\delta t))$ describes the importance weight, which are approximations to the posterior distribution of samples (filters) following (7a), then, the posterior (6a) can be indirectly approximated as the product between the importance distribution (7b) and the importance weight (7b),

$$\mathcal{P}(\Delta f(t)|\Delta p(t_0:t+\delta t);\Delta\omega) = \mathcal{Q}(\Delta f(t)|\Delta p(t_0:t+\delta t);\Delta\omega)\cdot\lambda(\Delta f(t_0:t+\delta t)) \quad (7a)$$

$$\lambda(\Delta f(t_0:t+\delta t)) = \frac{\mathcal{P}(\Delta f(t+\delta t)|\Delta p(t_0:t+\delta t);\Delta\omega)}{\mathcal{Q}(\Delta f(t+\delta t)|\Delta p(t_0:t+\delta t);\Delta\omega)} \quad (7b)$$

$$\mathcal{Q}(\Delta f(t+\delta t)|\Delta p(t_0:t+\delta t);\Delta\omega) = \mathcal{Q}(\Delta f(t+\delta t)|\Delta f(t_0:t);\Delta\omega)\cdot \mathcal{Q}(\Delta f(t)|\Delta p(t_0:t);\Delta\omega) \quad (7c)$$

By supposing the recursive type of the importance density and transferring (7c) into it, the recursive form of importance weight can be derived as,

$$\lambda(\Delta f(t_0:t)) = \lambda(\Delta f(t_0:t-\delta t))\cdot \mathcal{P}(\Delta p(t)|\Delta f(t_0:t);\Delta\omega)\left(\frac{\mathcal{P}(\Delta f(t)|\Delta f(t_0:t-\delta t);\Delta\omega)}{\mathcal{Q}(\Delta f(t)|\Delta f(t_0:t-\delta t);\Delta\omega)}\right) \quad (8a)$$

Let $\mathcal{Q}(\Delta f(t)|\Delta f(t_0:t-\delta t);\Delta\omega)$ equals to the value of transfer function $\mathcal{P}(\Delta f(t)|\Delta f(t_0:t-\delta t);\Delta\omega)$, then (8a) can be rewritten as,

$$\lambda(\Delta f(t_0:t)) = \lambda(\Delta f(t_0:t-\delta t))\cdot\mathcal{P}(\Delta p(t)|\Delta f(t_0:t);\Delta\omega) \quad (9a)$$

By combining (9a) and (7a), the posterior of the importance density can be directly derived. Subsequently, by updating the weight through (7b) and substituting Eqs. (7b) and (7c) into (7a), the posterior distribution regarding $\Delta f(t)$ in (5a) can be recursively solved.

III. Frequency constraint reformulation

This section examines frequency dynamic requirements within a union two-stage scheduling framework. On this basis, a revised formulation of frequency nadir restriction based on the proposed SFR model considering uncertain-variability features is extended, remaining frequency constraints including RoCoF and FCR can be found in [].

A. RoCoF constraint

RoCoF constraint aims to mitigate possible rapid changes of frequency derivation in the inertia response. Mathematically speaking, it is required that the differentiation of $\Delta f'(t)$ should not exceed the maximum rate of change of frequency, denoted as γ_{rocof}, *i.e.*, $\Delta f'(t) \le \gamma_{rocof}$. Since the maximum of $\Delta f'(t)$ equals to ${}^{\Delta p_{step}}/_{2H_\Sigma}$, then RoCoF constraint can be expressed as below,

$$0.5H_\Sigma^{-1}\Delta p_{step} \le \gamma_{rocof} \Leftrightarrow H_G + H_W \ge 0.5\times(\Delta p_{step}\gamma_{rocof}^{-1}) \quad (10)$$

B. FCR constraint

The restriction of FCR constraint is used to ensure sufficient frequency reserves provided by grid-connected generators that could provide frequency-supporting. By taking Eqs. (3a) and (3b) into consideration, the mathematical form of FCR constraint is expressed as follows,

$$\sum\nolimits_{g\in\mathcal{G}}(\overline{p}_g - p_{g,t}) \ge K_G R_G^{-1}\Delta f(t) - \alpha_2\int_0^t e^{\frac{t}{T_G}}\Delta f^{'}(t)dt, t\in(t_{IR},t_{PFR}] \quad (11a)$$

$$\sum\nolimits_{w\in\mathcal{W}_1}(\overline{p}_w - p_{w,t}) \ge [K_W R_W^{-1} + D_W]\Delta f(t+\delta t) - 2H_W\delta t\Delta f^{'}(t), t\in(t_{IR},t_{PFR}] \quad (11b)$$

$$\sum\nolimits_{w\in\mathcal{W}_2}(\overline{p}_w - p_{w,t}) \ge R_W^{-1}h_{nadir}, t\in(t_{IR},t_{PFR}] \quad (11c)$$

$$\sum\nolimits_{e\in\mathcal{E}}(\overline{p}_e - p_{e,t}) \ge R_E^{-1}h_{nadir}, t\in(t_{IR},t_{PFR}] \quad (11d)$$

where h_{nadir} denotes the maximum frequency deviation from the nominal frequency. From $t = t_{IR} \to t_{PFR}$, $\Delta f(t)$ should be kept within the range between h_{db} and h_{nadir}, and the change of $\Delta f^{'}(t) \to 0$ at $t = t_{PFR}$. Therefore, the upper bound of frequency containment reserves denoted by (11a) and (11b) can be approximated as,

$$\sum\nolimits_{g\in\mathcal{G}}\left(\overline{p}_g - p_{g,t}\right) \geq K_G R_G^{-1} h_{nadir}, t \in (t_{IR}, t_{PFR}] \tag{12a}$$

$$\sum\nolimits_{w\in\mathcal{W}_1}\left(\overline{p}_w - p_{w,t}\right) \geq [K_W R_W^{-1} + D_W] h_{nadir}, t \in (t_{IR}, t_{PFR}] \tag{12b}$$

C. $(Chance - constrained)\ nadir\ constraint$

The frequency nadir constraint aims to avoid $\varDelta f(t)$ beyond its predefined threshold $\varDelta f_{nadir}$ at the PFR process. Since the posterior of frequency derivation considering uncertain variability factors can be accurately solved through the Bayesian filter method, a chance-constrained frequency nadir constraint can be derived as follows,

$$\begin{gathered}\text{Prob}\left(\varDelta f(t) \leq \varDelta f_{nadir}\right) \geq 1 - \varepsilon_{SFR}; \forall t \in (t_{IR}, t_{PFR}] \\ \varDelta f(t) \propto \mathcal{P}(\varDelta f(t) | \varDelta p(t_0 : t); \varDelta\omega)\end{gathered} \tag{13}$$

$$\mathcal{P}\left(\varDelta f_{\text{nadir}} \leq \varDelta f_{\text{max}}\right) \geq 1 - \epsilon \tag{14}$$

where ε_{SFR} is the confidence coefficient associated with (13); $\varDelta f(t)$ can be generated from its posterior distribution denoted by (13). Using sampled average approximate (SAA) method, Eq. (13) can be equivalently reshaped as the deterministic form (**??**), which is listed below,

REFERENCES